3D-Dual Circular Metamaterial Structure

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***Abstract*— A 3D-dual circular metamaterial structure (3D-DCMS) is proposed to characterize the effective permittivity, effective permeability (effective parameters) and refractive index. The 3D-DCSRR consist of an outer ring, dual inner ring, which are connected by metallic vias. The structure is embedded to substrate which has a 0.1mm vertical distance to the dual inner ring. The proposed structure has a negative permeability at 2.075GHz. Therefore, an effective parameters Ɛ>0 and μ<0, resulting 3D-DCSRR as MNG metamaterial and the refractive index is negative.**

Keywords—Metamaterial, 2D metamaterial structures, 3D metamaterial structures.

I. INTRODUCTION

The word "Meta" is originated from Greek whose meaning is "beyond". "Metamaterials" has the extrinsic properties beyond the natural occurring materials. First metamaterial, exhibits the materials with negative permittivity and permeability are theoretically possible [1]. Later the term of metamaterial was synthesized in 1999, which was initially defined as "Macroscopic combinations having an artificial, three-dimensional, periodic cellular architecture aimed to produce an improved composites, not existing in nature, of two or more responses to specific excitation" [2]. The basic structure is split-ring resonator (SRR) introduced by pendry, he extended this study with resonators interacting with electromagnetic waves [3], and metallic wire [4].

Later by Smith and his group realize the LHM, then LHM with wire is to achieve the negative index refraction phenomenon [5]. The LHM based TL approaches are introduced independently and describe the metamaterial properties in backward electromagnetic waves [6-7] for 1D structure. Similarly, for 2D metamaterial structure properties are investigated [8]**.** Also, some other 2D metamaterial structure are investigated such as edge and broad side coupled resonator [9,10], axially symmetric SRR [11]. The concept of 2D/wire metamaterial structures are extended to 3D metamaterial structures. These structures are implemented with insert the metallic vias into the dielectric materials, results change the electromagnetic properties [12]. The excitation of 3D metamaterial are explored [13], 3D metamaterial with LTCC technique [14]. Recently, the large explosion of attention are given to 3D-metamaterial structures for various application.

The Proposed 3D- dual circular metamaterial structure is formed with outer ring, two dual inner rings, metallic vias and transmission lines. The structure is excited by closed boundary condition, when the axis of the ring is parallel to the magnetic field (H-field), the H-field produce a current in the 3D-DCMS, which gives increase to a strong magnetic resonance.

II. ANTENNA GEOMENTRY

The configuration of proposed structure is shown in "Fig.1". It is printed on RT-duroid substrate (ε_r=2.2) of dimension L×L and height (h). The 3D structure contains inner and dual outer rings of radius 'r_1' and 'r_2', width (d), gap (g), substrate height (h) and metallic vias height (h_1), airgap (h'). The outer rings of radius (r_1), transmission line are placed on bottom and top of the substrate, respectively. The two dual inner rings of same radius (r_2) are placed on top of metallic vias (h_1). The air gap (h') act as coupling gap between dual inner ring and transmission line for the 3D-dual circular metamaterial structure. The proposed structure is designed at resonant frequency of 2.075 GHz using EM simulator. All the dimensions of 3D-DCMS are listed in Table I.

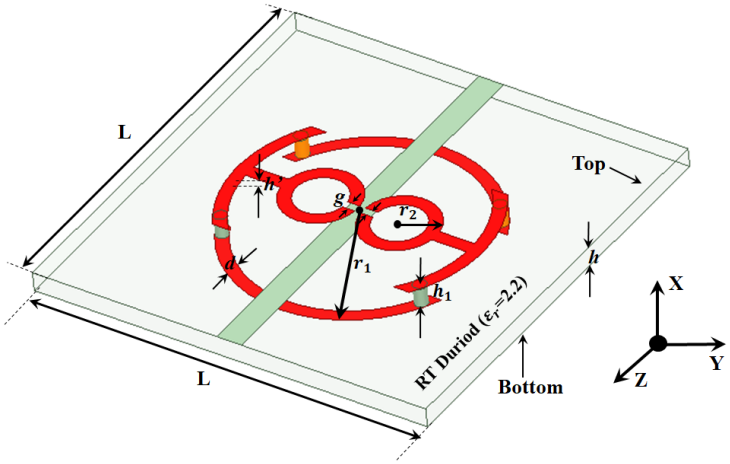


Fig. 1. Schematic view of 3D-dual circular metamaterial structure

TABLE I. DIMENSION OF 3D-DUAL CIRCULAR METAMATERIAL STRUCTURE (ALL DIMENSION IN MM)

L	r_1	r_2	d	h'	h	g
32	10	3	1	0.1	1.524	1

III. RESULTS AND DISCUSSION

"Fig. 2", shows the simulated Scattering-parameters (S_{11} & S_{21}) of 3D-DCMS metamaterial structure using EM simulation. It is observed that the structure shows the pass band characteristics. The transmission coefficient (S_{21}) has a peak notched and shows the good rejection level at resonant

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frequency (2.075 GHz). The reflection coefficient (S_{11}) attained almost zero at resonant frequency.

To examine the 3D-DCMS structure as metamaterial, the effective parameters are calculated are shown in "Fig. 3 (a) and (b)". Effective permittivity (ε_r) of the 3D-DCMS structure is shown in "Fig. 3 (a)". It exhibits positive value with spatial dispersion of low energy at the resonant frequency (2.075 GHz). The effective permeability (μ_r) of the 3D-DCMS structure is shown in "Fig. 3 (b)". It exhibits high negative peak value at the resonant frequency. Therefore, a strong magnetic response is produce for negative permeability. The effective parameters $\varepsilon>0$ and $\mu<0$, resulting 3D-DCMS as MNG metamaterial. The refractive index for 3D-DCMS is negative as shown in "Fig. 4". It defines the impedance of 3D-DCMS structure is matched. Hence, the coupling is strong with electromagnetic waves.

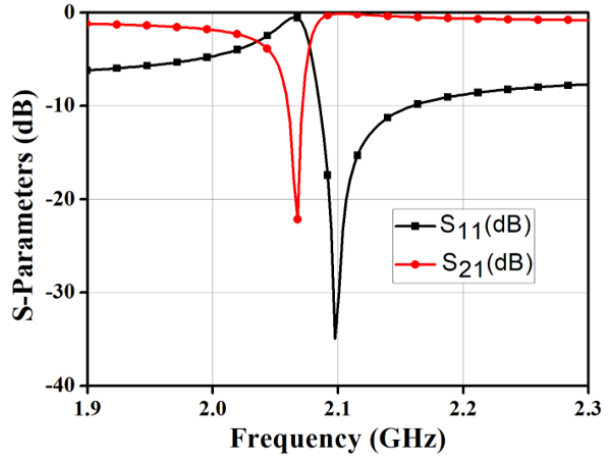
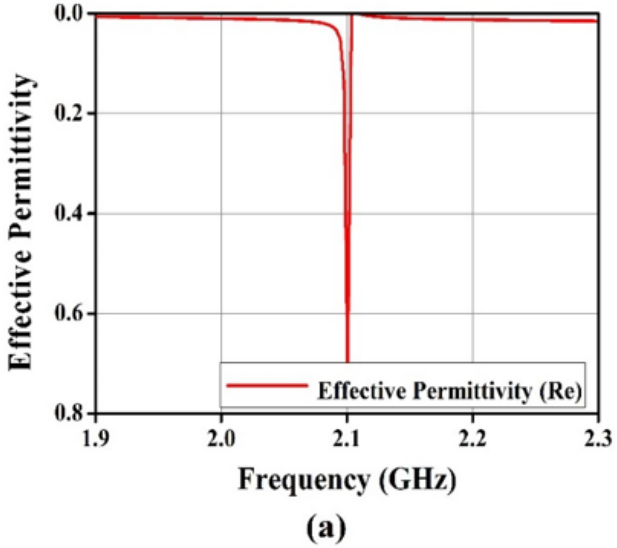


Fig. 2. Simulated scattering-parameters of 3D-dual circular metamaterial structure



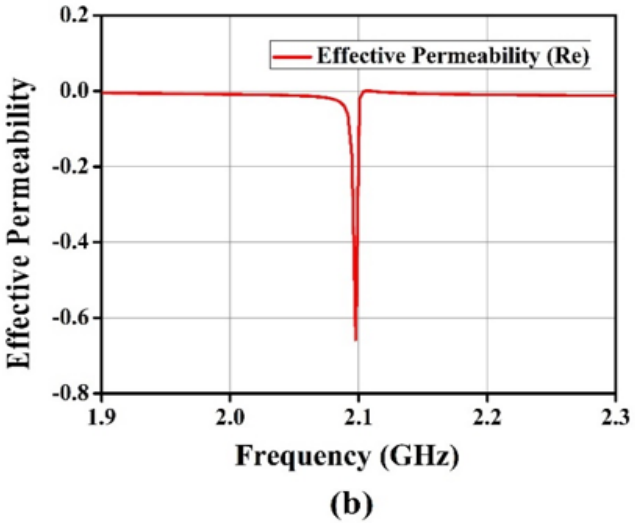


Fig. 3. Extracted effective parameters(real) vs frequency: (a) permittivity, (b) permeability

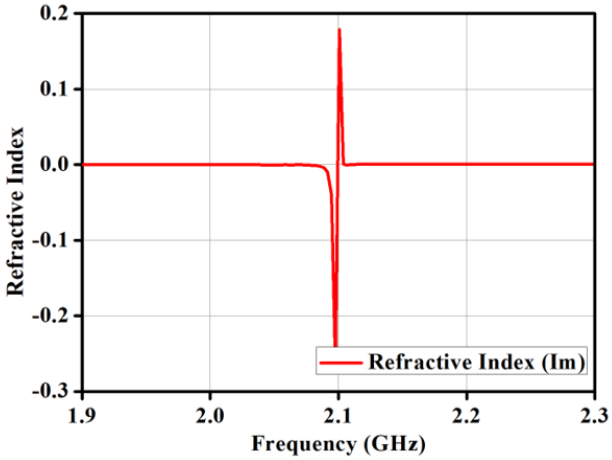


Fig. 4. Refractive index vs frequency (Imaginary) of 3D-DCMS

The surface current distribution is shown in "Fig.5". We assume, an electric (E) field is applied corresponding to the axis of the dual inner ring (r_2), current is induced and produce magnetic field. There is strong coupling between transmission line and dual inner rings through the air gap. The current is splitted and flow dual inner ring (r_2) to outer ring (r_1) through metallic vias in both anti-clockwise and clockwise direction.

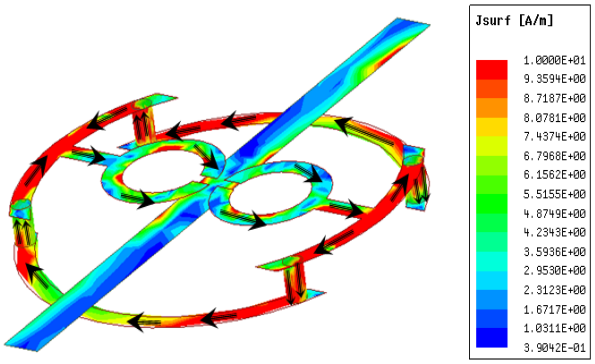


Fig. 5. Surface current density of 3D-DCMS structure with current distribution

IV. CONCLUSION

The 3D dual circular metamaterial structure are analysed using Scattering-parameters to calculate effective permittivity, effective permeability and refractive index to analyze structure as metamaterial structure. The conducted analysis for 3D metamaterial structures exhibits negative material characteristics that are suitable for various high frequency wireless applications. In addition, the 3D metamaterial structure used as metamaterial antennas.

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